



Confusion Matrix Explained: Calculating Accuracy, TPR, FPR, TNR, Precision, and Prevalence



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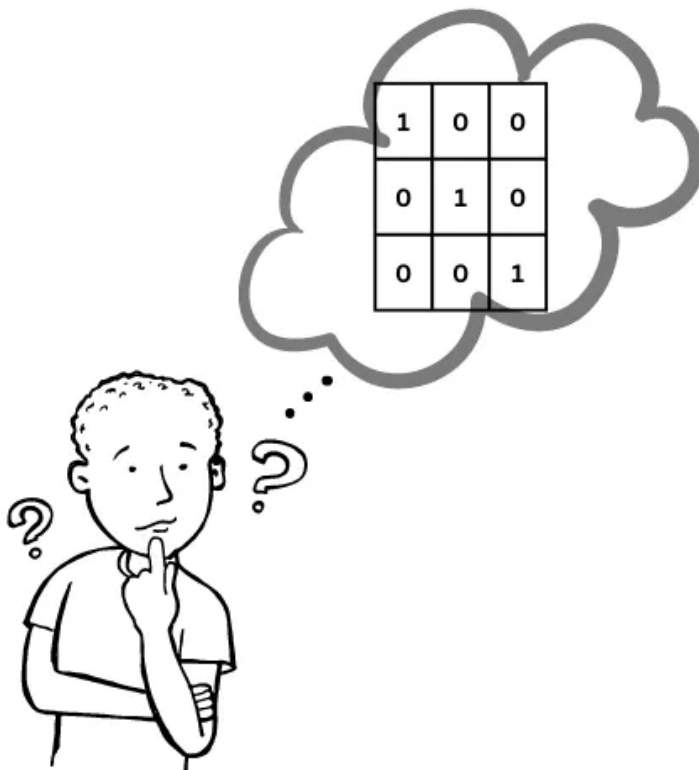


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In this blog we are going to study one of the most useful performance evaluation matrix for classification problem. Confusion matrix is the easiest way to measure the performance of a classification problem where the output can be of two or more type of classes. Evaluation of the performance of a classification model is based on the counts of test records correctly and incorrectly predicted by the model. These counts are tabulated in a table known as a confusion matrix. Each entry f_{ij} indictable denotes the number of records for class i predicted to be of class j .

A confusion matrix is nothing but a table with two dimensions viz. “Actual” and “Predicted” and furthermore, both the dimensions have “True Positives (TP)”, “True Negatives (TN)”, “False Positives (FP)”, “False Negatives (FN)”. The confusion matrix is a matrix used to determine the performance of the classification models for a given set of test data.

It can only be determined if the true values for test data are known. Since it shows the errors in the model performance in the form of a matrix, hence also known as an error matrix. The matrix is divided into two dimensions, that are predicted values and actual values along with the total number of predictions. Predicted values are those values, which are predicted by the model, and actual values are the true values for the given observations.

It evaluates the performance of the classification models, when they make predictions on test data, and tells how good our classification model is.

At times, it is difficult to judge the accuracy of a model by just looking at the accuracy because of problems like unequal distribution. So, a better way to check how good your model is, is to use a confusion matrix.

First, let's look at some key terms.

Classification accuracy — This is the ratio of the number of correct predictions to the number of predictions made.

True positives — Correct predictions of true events.

False positives — Incorrect predictions of true events

True negatives — Correct predictions of false events

False negatives — Incorrect predictions of false events.

The confusion matrix is now simply a matrix containing true positives, false positives, true negatives, false negatives.

		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

Let's understand this concept with example.

		Predicted	
		P	N
Actual	P	TP	FN
	N	FP	TN

		Predicted	
		P	N
Actual	P	100	5
	N	10	50

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There are six evaluation metrics that can be computed in a confusion matrix. It is